

DSA210 – Introduction to Data Science (Fall 2025–2026)

GameTasteDSA210: A Personal Video-Game Recommender using RAWG Metadata

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Abstract

This project builds a single-user, content-based recommendation pipeline for video games by combining my manually recorded ratings (1–10 scale) with public metadata from the RAWG platform (genres, tags, Metacritic score, release year, and popularity). Using exploratory analysis, hypothesis tests, and supervised regression models, I learn interpretable preference signals and then predict my probable ratings for thousands of unrated games. The final recommender produces 17 recommendations with a healthy mix of mainstream and less obvious (indie/older) titles, including *Dwarf Fortress*, which strongly aligns with my known preference for *RimWorld* despite *RimWorld* not appearing in my dataset.

1 Motivation

I wanted to test whether a recommender system built from only my own ratings can (1) discover consistent preference patterns and (2) generate genuinely useful next-game suggestions rather than only obvious mainstream picks. In practice, I often enjoy games for specific structural reasons (e.g., systemic depth, progression loops, strong atmosphere), so I was curious whether high-level metadata such as genres/tags can capture these signals well enough to recommend games I would actually want to play next.

2 Data Source and Collection

2.1 Personal Ratings (Private Dataset)

I used a personal list of 256 games that I rated manually on a 1–10 scale. This data represents my taste over multiple years, with no year limit imposed during collection. The ratings are subjective and reflect my personal preferences.

2.2 RAWG Metadata (Public Dataset)

For large-scale candidate games, I collected a RAWG dataset filtered to entries with available Metacritic scores (yielding 7,045 games). For each game, I used the fields: **genres**, **tags**, **metacritic**, **released__year**, and (for filtering/robustness) **ratings__count**.

2.3 Merged Modeling Dataset

After name normalization and merging, 229 of my rated games matched the RAWG metadata and were used for analysis/modeling. The remaining RAWG games (6,816) form the unrated candidate pool for recommendation.

Table 1: Dataset sizes used in the pipeline

Component	Size
My rated games (raw list)	256
RAWG candidate set (Metacritic available)	7,045
Matched rated games used for modeling	229
Unrated candidate games scored by the model	6,816
Final recommendations (after popularity filter)	17

3 Data Analysis Pipeline

3.1 Stage 1: Feature Engineering and Preference Signals

Each game is represented by:

- One-hot encoded **genre** and **tag** indicators (MultiLabelBinarizer).
- Numeric features: **release year**, **Metacritic**, and **$\log(1+\text{ratings_count})$** (in the recommender).

To reduce noise, infrequent tags/genres were filtered in the descriptive analysis, and rare features were dropped prior to some modeling steps.

3.2 Stage 2: Exploratory Data Analysis

I analyzed the distribution of my ratings and how they relate to release year and external quality signals (Metacritic).

Key descriptive stats over the 229 matched games:

Mean = 7.22, Median = 7.00, Std = 1.32, Skewness = -0.16 .

Metacritic has a moderate positive relationship with my rating (Pearson $r \approx 0.40$), while release year shows almost no linear relationship ($r \approx -0.02$).

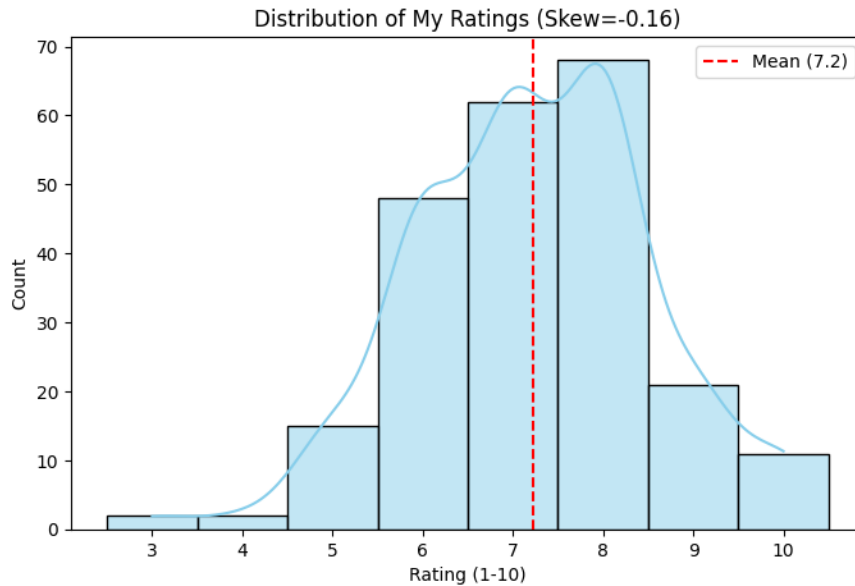


Figure 1: Distribution of my ratings (matched games).

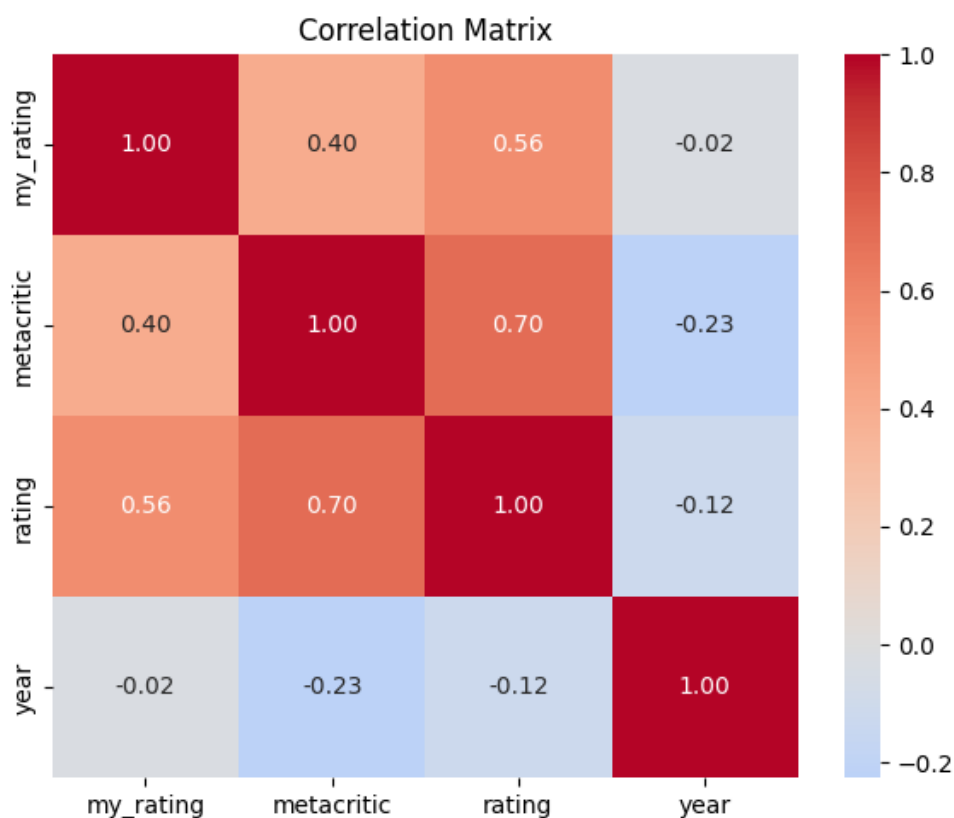


Figure 2: Correlation matrix among key numeric variables (my rating, Metacritic, year).

3.3 Stage 3: Hypothesis Testing on Tags/Genres

To identify statistically meaningful preference signals, I used Mann–Whitney U tests comparing ratings of games *with* vs *without* each feature (only features appearing in at least 5 games). For exploratory purposes, I considered $p < 0.10$ as “significant” signals (noting multiple-testing limitations).

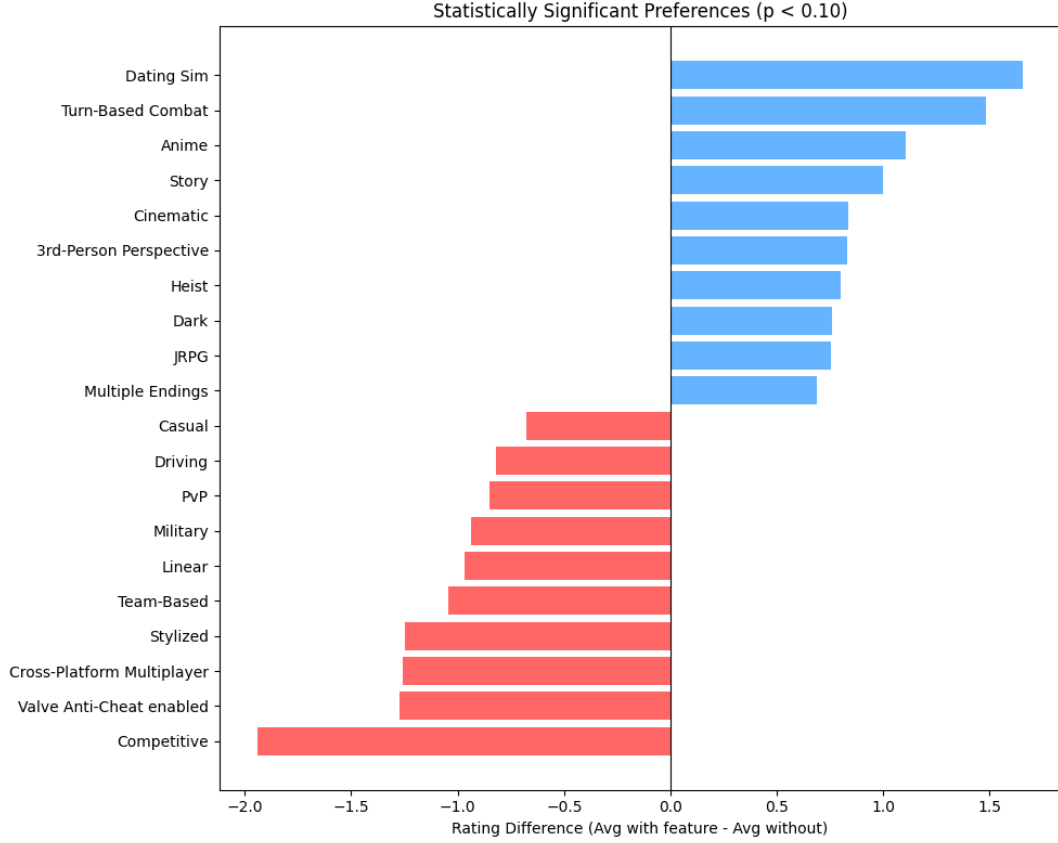


Figure 3: Statistically significant preferences (Mann–Whitney U, $p < 0.10$).

3.4 Stage 4: Supervised Taste Modeling and Evaluation

I evaluated three regressors on a train/test split (75/25) of the matched rated games:

- Ridge regression (L2-regularized linear model)
- Lasso regression (L1-regularized model with feature selection)
- Random Forest regression (non-linear ensemble)

Table 2: Model performance on held-out test set (229 matched rated games).

Model	MAE	RMSE	R^2
Ridge (L2)	1.25	1.65	-0.80
Lasso (L1)	0.91	1.17	0.09
Random Forest	0.90	1.16	0.11

The absolute errors are around ~ 0.9 rating points for the best models, but R^2 is low, consistent with a small, noisy, single-user dataset where many preferences are not perfectly explained by metadata.

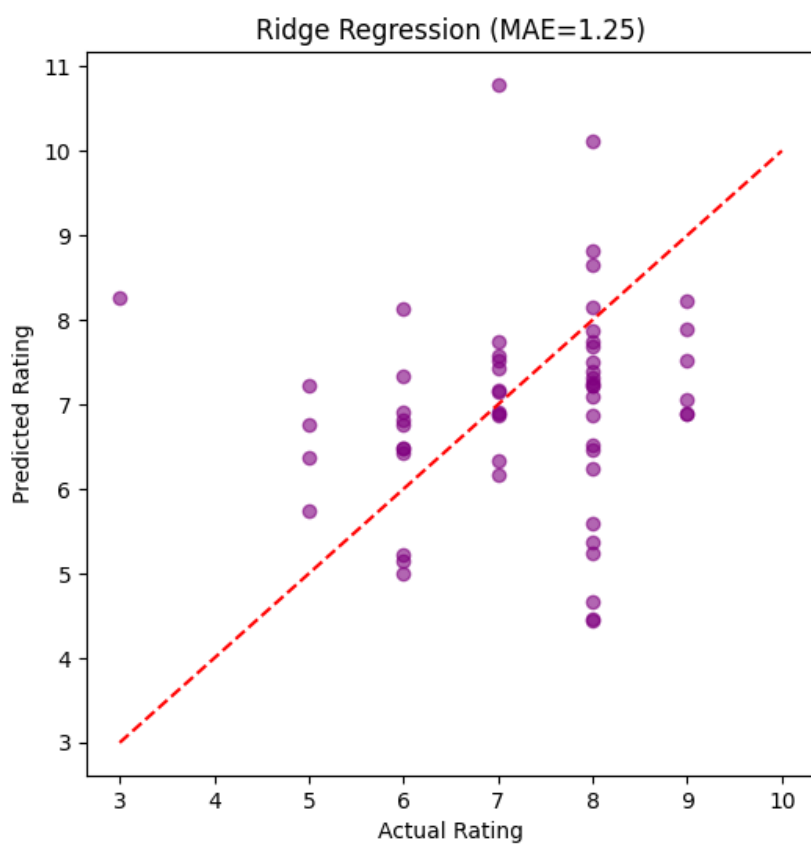


Figure 4: Ridge: actual vs predicted ratings on the test split.

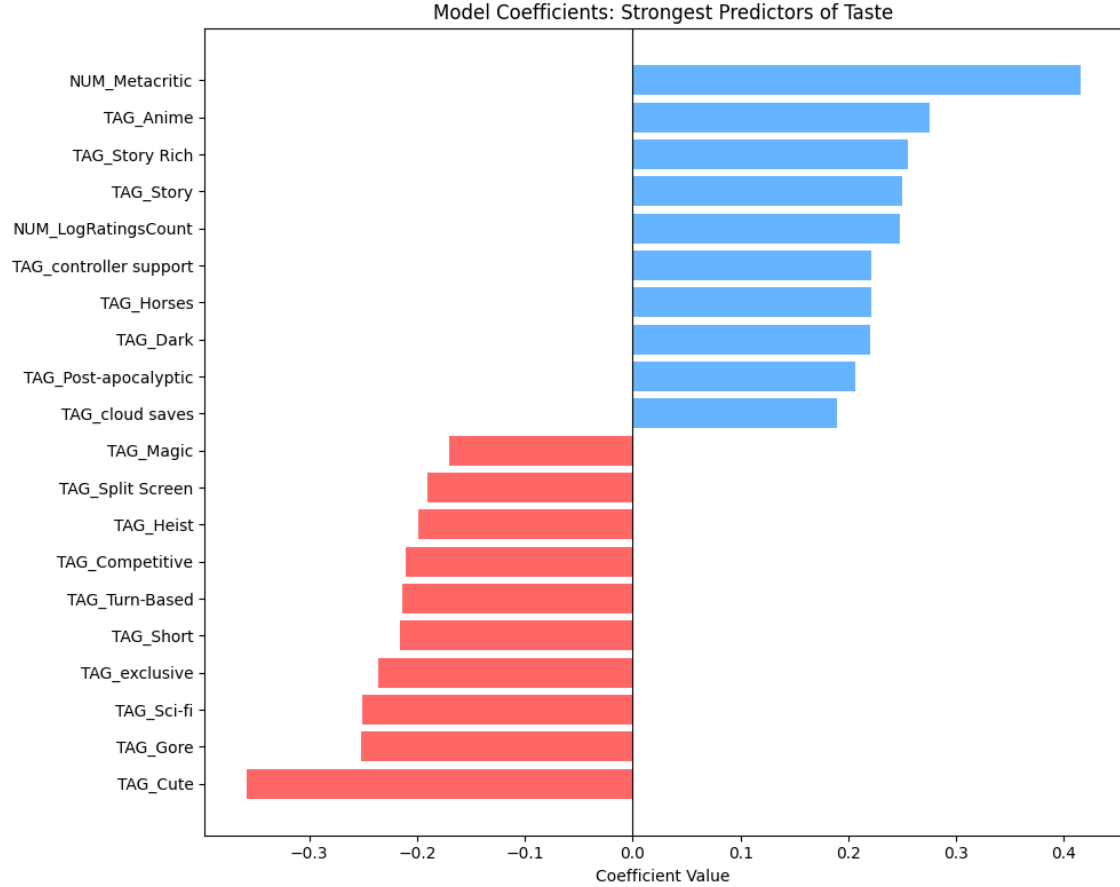


Figure 5: Ridge coefficients: strongest positive/negative taste predictors (interpretable signals).

3.5 Stage 5: Recommendation Generation

For recommendations, I trained a Random Forest regressor on all 229 matched rated games, predicted ratings for 6,816 unrated candidates, and recommended only games with more than 50 community ratings to avoid extremely obscure titles with unreliable metadata. This resulted in 17 final recommendations.

4 Findings

4.1 What I tend to like (effect-size / lift analysis)

A simple, interpretable analysis is the *rating lift*:

$$\text{lift}(\text{feature}) = E[\text{rating} \mid \text{feature}] - E[\text{rating}].$$

Across genres, **RPG** stands out as the clearest positive signal, while many other genres are slightly negative or near-neutral.

Table 3: Top genre signals by rating lift (matched games).

Genre feature	Count	Lift
GENRE: RPG	66	0.38
GENRE: Action	170	0.06
GENRE: Platformer	16	0.03
GENRE: Adventure	74	-0.06
GENRE: Puzzle	14	-0.15

Tags provide sharper signals. For example, *Anime*, *Cinematic*, and *Detective* are associated with higher-than-average ratings, while strongly competitive/multiplayer-oriented tags tend to be negative in my data.

Table 4: Example tag signals by rating lift (matched games).

Tag feature	Count	Lift
TAG: Anime	11	1.05
TAG: 3rd Person Perspective	14	0.78
TAG: Heist	7	0.78
TAG: Detective	8	0.78
TAG: Cinematic	16	0.78
TAG: Valve Anti Cheat enabled	9	-1.22
TAG: Team Based	9	-1.00
TAG: Linear	7	-0.94
TAG: Military	12	-0.89
TAG: Mystery	8	-0.85

4.2 Hypothesis-test highlights

Using Mann–Whitney U tests, I found multiple statistically significant features. A few large-effect examples are shown below.

Table 5: Examples of statistically significant features (Mann–Whitney U, $p < 0.10$).

Feature	Count	Mean difference
Dating Sim	6	1.65
Turn-Based Combat	6	1.48
Anime	11	1.10
Story	5	1.00
Cinematic	16	0.84
Competitive	6	-1.94
Valve Anti-Cheat enabled	9	-1.27
Cross-Platform Multiplayer	6	-1.26
Stylized	5	-1.25
Team-Based	9	-1.04

4.3 Recommendation quality (qualitative evaluation)

The recommender performed well in practice: it did not only output the most mainstream obvious games, but suggested a balanced mix, including older classics and less-discussed games. Most of the recommended titles overlapped with games I already wanted to play next, which suggests the model learned a meaningful personal representation of my taste.

A particularly interesting outcome is that the recommender suggested *Dwarf Fortress* with a high predicted score. *Dwarf Fortress* is a well-known predecessor and close relative of *RimWorld*, a game I strongly love and would likely rate 10/10. However, *RimWorld* was not in my rated dataset. This indicates that the recommender inferred a preference for the underlying *type* of game (systemic simulation/sandbox depth) using only metadata correlations rather than direct memorization.

Table 6: Top recommendations (predicted rating) from the final list.

Game	Pred. rating	Year	Genres
Okami	8.87	2006	Action Adventure
Maniac Mansion: Day of the Tentacle	8.80	1993	Adventure Puzzle
NieR:Automata Game of the YoRHa Edition	8.76	2019	Action
Final Fantasy XII	8.72	2006	RPG
The Legend of Zelda: Ocarina of Time 3D	8.71	2011	Adventure RPG Puzzle
BioShock Infinite	8.69	2013	Action Shooter
Metroid Prime: Trilogy	8.68	2009	Action Adventure
Paper Mario (2000)	8.60	2000	RPG
Resident Evil Code: Veronica	8.59	2000	Action Adventure
Dwarf Fortress	8.58	2006	RPG Strategy Simulation Indie

5 Limitations and Future Work

5.1 Limitations

- **Single-user and small labeled set:** Only 229 matched rated games were available for supervised learning, which limits generalization and makes metrics noisy.
- **Subjective labels:** Ratings are personal and may be inconsistent across time and context.
- **Metadata-only representation:** Genres/tags cannot fully encode gameplay feel (e.g., pacing, difficulty curve, UI/controls, narrative quality).
- **Selection bias:** The RAWG candidate set was filtered to games with Metacritic values, which biases the pool toward more documented/popular titles.
- **Limited evaluation:** While train/test metrics were computed, recommendation quality is primarily evaluated qualitatively (whether I want to play the results).

5.2 Future Work

- Add **multi-user** data and explore collaborative filtering (user-item matrix factorization).

- Use richer representations (e.g., **text embeddings** from descriptions/reviews) to capture nuance beyond tags.
- Model **temporal preference drift** (ratings collected over years; tastes change).
- Incorporate additional signals (platform, playtime, completion status, price/discount patterns, etc.).
- Add a structured evaluation protocol (e.g., offline ranking metrics + a controlled “play next” validation set).

6 Reproducibility

All scripts, data files, and generated outputs are included in the accompanying GitHub repository. The analysis is fully reproducible by installing dependencies (see `requirements.txt`) and running the objective scripts that generate the figures and CSV outputs used in this report.

References

- [1] RAWG Video Games Database, API and metadata. <https://rawg.io/apidocs>